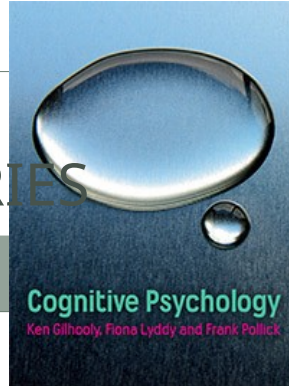


DUAL PROCESS THEORIES



OTHER APPROACH: DUAL-PROCESS THEORIES

- Different forms and different theories
- Same background idea:
 - There are two different cognitive systems:
 - Quick, emotional, irrational, automatic, also with animals: *"in service of the genes"*
 - Slow, conscious, logical, analytical, only humans

OTHER APPROACH: DUAL-PROCESS THEORIES

- Popular (also in the past)
 - Freud, ...
- Also outside science:
 - Star Trek

Dual Process theories	System 1	System 2
Evans (1984; 1989) Pollock (1991) Reber (1993) Epstein (1994) Levinson (1995) Evans and Over (1996) Hammond (1996) Sloman (1996) Klein (1998)	Heuristic processing Quick and inflexible modules Implicit cognition Experiential system Interactional Intelligence Tacit thought processes Intuitive Cognition Associative system Recognition Primed Decision	Analytic processing Intellection Explicit learning Analytic intelligence Analytic Intelligence Explicit thought processes Analytical cognition Rule-based system Rational Choice Strategy

STAR TREK

- Mr. Spock
- Half human, half Vulcan
- No emotions, only logical thinking



STAR TREK

- Dr. McCoy, alias "Bones"
- Antipole Spock
- Full of emotions, often irrational, ...



HEURISTIC SYSTEM, SYSTEM 1, TASS (THE **AUTONOMOUS SET OF SYSTEMS**)

- Universal cognition (human and animal)
- Not 1 system, but set of subsystems
 - Instinctive behaviors (innately programmed)
 - Associative learning (neural networks)
 - Rapid, parallel, automatic
 - Only final product is posted in consciousness

ANALYTIC SYSTEM, SYSTEM 2

- Evolved much more recently
- Uniquely human
- Slow, sequential in nature
- Makes use of central working memory system
- Permits abstract hypothetical thinking

BELIEF-BIAS EFFECT

- Experiments: Participants instructed to treat problems as logical reasoning task
- However: Consistent influence of prior believability of conclusion
- 4 syllogisms as examples (see following slides)

VALID ARGUMENT, BELIEVABLE CONCLUSION: NO CONFLICT

- *No police dogs are vicious.*
- *Some highly trained dogs are vicious.*
- *Therefore, some highly trained dogs are not police dogs.*
- *PolDo*
- *PolDo*
- *PolDo*
- _____
- *ViDo*
- *ViDo*
- *ViDo*

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- *No police dogs are vicious.*
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- *ViDo = HTDo*
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- _____
- ViDo = HTDo ViDo = HTDo ViDo = HTDo
- ~~ViDo = HTDo ViDo = HTDo ViDo = HTDo~~
- ViDo ViDo ViDo

Irrespective of construction of Multiple Models (or counterexamples), reasoners will come up with valid conclusion

VALID ARGUMENT, UNBELIEVABLE CONCLUSION: CONFLICT

- No nutritional things are inexpensive.
- Some vitamin tablets are inexpensive.
- Therefore, some vitamin tablets are not nutritional.

- NuTh NuTh = VitT NuTh = VitT
- NuTh NuTh = VitT NuTh = VitT
- NuTh NuTh = VitT NuTh
- _____
- InExp = VitT InExp = VitT InExp = VitT
- InExp = VitT InExp = VitT InExp = VitT
- InExp InExp InExp

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- ~~InExp = VitT InExp = VitT InExp = VitT~~
- InExp InExp InExp

Irrespective of construction of Multiple Models (or counterexamples), reasoners will come up with valid conclusion
But initial model might be influenced by content

INVALID ARGUMENT, BELIEVABLE CONCLUSION: CONFLICT

- No addictive things are inexpensive.
- Some cigarettes are inexpensive.
- Therefore, some addictive things are not cigarettes.

- AT AT = Cig AT = Cig
- AT AT = Cig AT = Cig
- AT AT = Cig AT
- ~~InExp = Cig InExp = Cig InExp = Cig~~
- ~~InExp = Cig InExp = Cig InExp = Cig~~
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• AT	AT = Cig	AT = Cig
• AT	AT = Cig	AT
•		
• InExp = Cig	InExp = Cig	InExp = Cig
• InExp = Cig	InExp = Cig	InExp = Cig
• InExp	InExp	InExp

INVALID ARGUMENT, BELIEVABLE CONCLUSION: CONFLICT

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• AT	AT = Cig	AT = Cig?
• AT	AT = Cig	AT
•		
• InExp = Cig	InExp = Cig	InExp = Cig
• InExp = Cig	InExp = Cig	InExp = Cig
• InExp	InExp	InExp

If I do not construct the middle counterexample, I will conclude that the given conclusion is ok:

- That's in line with beliefs
- That's not in line with totally fleshed out models

INVALID ARGUMENT, UNBELIEVABLE CONCLUSION: NO CONFLICT

- No millionaires are hard workers.
- Some rich people are hard workers.
- Therefore, **some millionaires are not rich people.**

• Mil	Mil = RiPe	Mil = RiPe
• Mil	Mil = RiPe	Mil = RiPe
• Mil	Mil = RiPe	Mil
•		
• HaWo = RiPe	HaWo = RiPe	HaWo = RiPe
• HaWo = RiPe	HaWo = RiPe	HaWo = RiPe
• HaWo	HaWo	HaWo

INVALID ARGUMENT, UNBELIEVABLE CONCLUSION: NO CONFLICT

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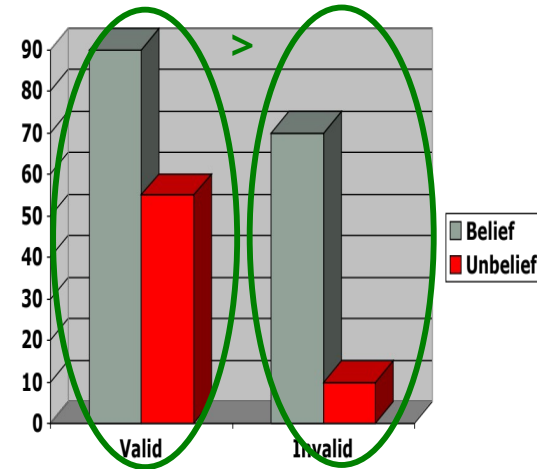
• Mil	Mil = RiPe	Mil = RiPe
• Mil	Mil = RiPe	Mil = RiPe
• Mil	Mil = RiPe	Mil
•		
• HaWo = RiPe	HaWo = RiPe	HaWo = RiPe
• HaWo = RiPe	HaWo = RiPe	HaWo = RiPe
• HaWo	HaWo	HaWo

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 • _____
 • HaWo = RiPe HaWo = RiPe HaWo = RiPe
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 • HaWo HaWo HaWo

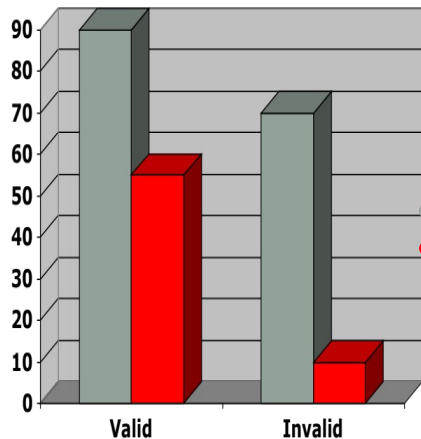
PERCENTAGES ACCEPTANCE OF CONCLUSIONS



Influence of 2 factors:

- More acceptance when valid

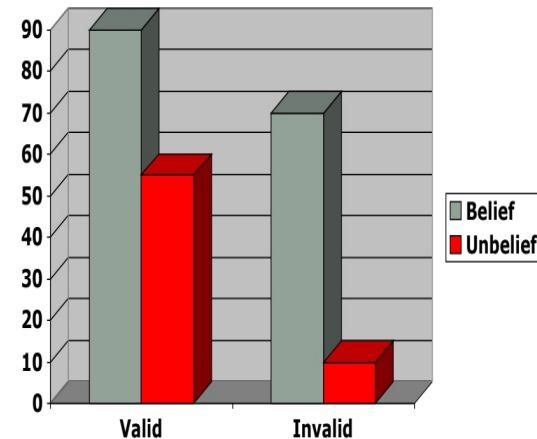
PERCENTAGES ACCEPTANCE OF CONCLUSIONS



Influence of 2 factors:

- More acceptance when valid
- More acceptance when believable

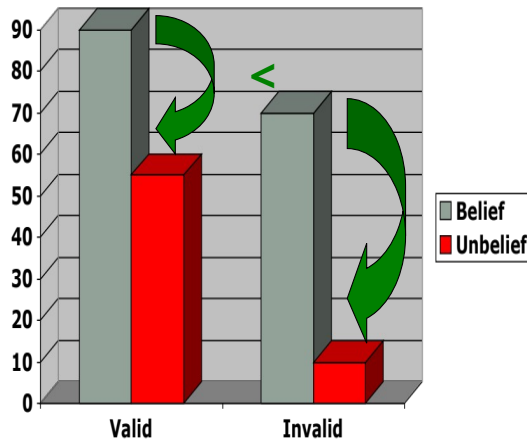
PERCENTAGES ACCEPTANCE OF CONCLUSIONS



Influence of 2 factors:

- More acceptance when valid
- More acceptance when believable
- Bigger effect of believability on invalid problems

PERCENTAGES ACCEPTANCE OF CONCLUSIONS



Influence of 2 factors:

- More acceptance when valid
- More acceptance when believable
- Bigger effect of believability on invalid problems

EXPLANATION OF EFFECTS IN GENERAL TERMS

- Participants attempt to reason logically
- However:
 - Influence of prior beliefs is extremely difficult to suppress
- Therefore:
 - Beliefs compete for control of the responses made

EXPLANATION OF EFFECTS IN DPT-TERMS

- Influence of prior beliefs= System 1
- Logical thinking = System 2
- Ability to resolve the conflicts in favour of logic
 - Correlated with measures of intelligence
 - Declines sharply with age (WM declines with age, WM is related to System 2)

THE WASON SELECTION TASK

Here are 4 cards. Each of them has a letter on one side and a number on the other side.

Two of these cards are shown with the letter side up, and two with the number side up.



Indicate which of these cards you have to turn over in order to determine whether the following claim is true:
If a card has a vowel on one side, then it has an odd number on the other side.

THE BEER/COKE TASK

In its cracking down against drunk drivers, New York law enforcement officials are revoking liquor licences left and right. You are a bouncer in a Boston Bar, and you'll lose your job unless you enforce the following law:

If a person is drinking beer, then he must be over 20 years old.

The cards below have info about 4 people sitting at a table in your bar. Each card represents one person. One side of the card tells what a person is drinking, the other side the person's age. Indicate only those card(s) you definitely need to turn over to see if any of these people are breaking the law.

Drinking
beer

Drinking
Coke

25 years old

16 years old

TEST OF MATCHING BIAS

- Abstract version with negations:
 - *If A then not-2.*
- Suddenly more logically correct answers: A & 2
 - Not because of better analytical reasoning (= System 2)
 - Because of System 1 = matching-bias
- Hence, matching bias sometimes good, sometimes bad

HENCE ...

- Strong influence of content.
- Abstract version:
 - System 2 must solve the task
 - Danger: System 1:
 - Matching bias
 - = use/select what is mentioned in rule
- Concrete version:
 - System 1 gives the correct answer on basis of knowledge of situation

WST AND INDIVIDUAL DIFFERENCES

- System 2 requires capacity (WM)
 - $WM \approx \text{Intelligence}$
- Hypothesis:
 - System 2 is related to IQ, System 1 not
- Evidence:
 - More normative correct responses if high in cognitive ability (SAT)
 - Much sharper differences in SAT-scores between solvers and non-solvers of the abstract problem than the concrete problem

DPT AND MMT

- DPT and MMT not incompatible
- Johnson-Laird & Khemlani, 2014:
- *...the theory postulates two separate systems for reasoning, one for intuitions and one for deliberations — a familiar distinction in “dual process” theories of reasoning (see, e.g. Evans, 2008; Kahneman, 2011; Reitman, 1965; Sloman, 1996; Stanovich, 1999; Verschueren, Schaeken, & d’Ydewalle, 2005)*
- *The model theory distinguishes between the two systems in computational power, and we have implemented both of them in computer programs (Khemlani & Johnson-Laird, 2012a; Khemlani, Lotstein, & Johnson-Laird, 2012).*

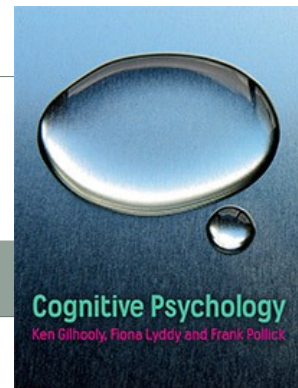
DPT AND MMT

- *In contrast, the deliberative system, which is sometimes known as “system 2”, has access to working memory, and so it can search for alternative mental models, and carry out recursive processes, such as counting and arithmetical operations, until they overwhelm its processing capacity.*

DPT AND MMT

- *The intuitive system, which is sometimes known as “system 1”, has no access to working memory, and so it can represent only one mental model at a time (Johnson-Laird, 1983, chap. 6), and it cannot carry out recursive processes, including arithmetical operations such as counting.*
- *It lacks even the computational power of a finite-state automaton (Hopcroft & Ullman, 1979) because it can carry out a loop of operations for only a small finite number of times—a restriction that is built into its computer implementation.*

2. DECISION MAKING



2. DECISION MAKING

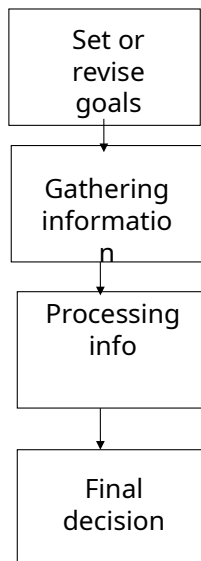
INTRODUCTION



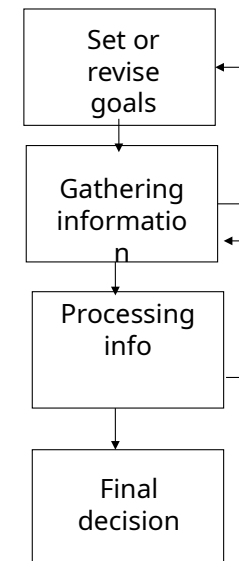
INTRODUCTION

- Decision making
 - choosing between alternative options/actions
 - weighing up consequences.
- Easy in principle

1. DECISION MAKING: SIMPLE, ISN'T IT 🌴



1. DECISION MAKING: SIMPLE, ISN'T IT 🌴



1. DECISION MAKING: SIMPLE, ISN'T IT 🌴

- Maximise (subjective) expected utility (SEU)
 - Subjective – not necessarily objective
 - Expected – is about future
 - Utility = value for a person (not always in terms of cash)
- $SEU = \sum(p_i \times u_i)$

1. DECISION MAKING: SOMEWHAT SIMPLE, ISN'T IT 🌴

- Maximise (subjective) expected utility (SEU)
 - Subjective – not necessarily objective
 - Expected – is about future
 - Utility = value for a person (not always in terms of cash)
- $SEU = \sum(p_i \times u_i)$

6 STEPS IN DECISION MAKING

1. Breaking decision (*which master?*) into independent dimensions
 - (practical, content, money)
2. Determining the relative weights
 - (vb practical more important than money)
3. Listing alternatives
 - (A, B, C, D)
4. Ranking alternatives along the dimensions
5. Multiplying ranking by weighting
6. Choosing alternative with the highest value

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D
practical				
content				
money				

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D
<i>practical</i>	6	5	9	1
<i>content</i>				
<i>money</i>				

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D
<i>practical</i>	6	5	9	1
<i>content</i>	8	5	4	7
<i>money</i>				

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D
<i>practical</i>	6	5	9	1
<i>content</i>	8	5	4	7
<i>money</i>	7	9	5	8

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D
<i>practical</i>	6	5	9	1
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<i>money</i>	7	9	5	8
U =	21	19	18	16

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D
<i>practical</i>	6	5	9	1
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U =	<u>21</u>	19	18	16

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D	Weight
<i>practical</i>	6	5	9	1	
<i>content</i>	8	5	4	7	
<i>money</i>	7	9	5	8	
U =	21	19	18	16	

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D	Weight
<i>practical</i>	6	5	9	1	3
<i>content</i>	8	5	4	7	1
<i>money</i>	7	9	5	8	2
U =	21	19	18	16	

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D	Weight
<i>practical</i>	6	5	9	1	3
<i>content</i>	8	5	4	7	1
<i>money</i>	7	9	5	8	2
U =	21	19	18	16	
	40	39	41	26	

6 STEPS IN DECISION MAKING

- Ex: which master?

	A	B	C	D	Weight
<i>practical</i>	6	5	9	1	3
<i>content</i>	8	5	4	7	1
<i>money</i>	7	9	5	8	2
U =	21	19	18	16	
	40	39	<u>41</u>	26	

RESEARCH

- = Normative Approach
 - best solution.
- But, people are not doing this!
- Descriptive Approach
 - how decisions take place
 - how they are made
- Systematic, frequent biases
 - = ways of thinking/tackling problems which lead to systematic errors
- Some problems/examples

2. DECISION MAKING

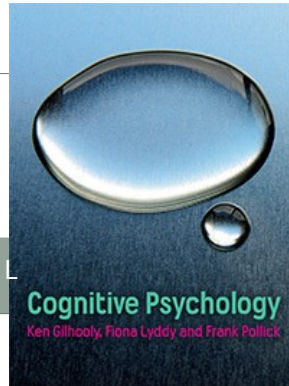
SOME DM-PROBLEMS



- These slides will only be posted on Toledo after today's lecture.

2. DECISION MAKING

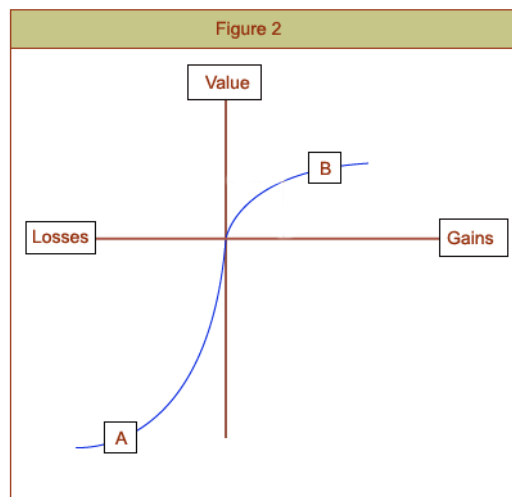
SOME HEURISTICS IN MORE DETAIL



WHICH CHOICE DO YOU PREFER?

- Choice A: You take a bet in which a fair coin is tossed.
 - If the toss comes up heads, you win \$10.
 - If the toss comes up tails, you lose \$10.
- Choice B: You don't take the bet.
- Rational prediction (from expected utility theory):
 - there should be no strong preference for either choice
 - because each choice has an average expected value of \$0
- Results
 - people strongly prefer Choice B
- Why?
 - losses loom larger than gains
 - a single monetary value has different utilities depending on whether it's a loss or gain
 - losing a particular amount of money is seen as more of a loss than winning it is seen as a gain
 - again, utilities vary with the context in which they are perceived

LOSSES LOOM LARGER THAN GAINS



IN AN EPIDEMIC, WHICH OF THE FOLLOWING TWO CHOICES DO YOU PREFER?

- CHOICE 1:
 - 80% chance of losing 100 lives
- CHOICE 2:
 - 100% chance of losing 75 lives
- CHOICE 3:
 - 8% chance of losing 100 lives
- CHOICE 4:
 - 10% chance of losing 75 lives
- Rational:
 - The 2 questions are formally identical
 - Choices 3 & 4 = choices 1 & 2 divided by a factor of 10
 - Hence, the relative values of choices 1 and 2 must be the same as the relative values of choices 3 and 4
 - Hence, if you preferred 1 over 2, you should have also preferred 3 over 4
- Results:
 - Most people prefer 1 over 2, but 4 over 3
- Explanation:
 - People don't like the certainty of choice 2, 100%
 - When there is no certainty, the lower lives lost (75) is more attractive, and the slightly higher probability (10%) is ignored

BASE-RATE NEGLECT: DISEASE Y PROBLEM

- Given to Staff and 4-year Harvard Medical School
- If a test, to detect a disease whose prevalence is 1/1000, has a false positive rate of 5%, what is the chance that a person found to have a positive result actually has the disease, assuming that you know nothing about the person's symptoms or signs? ----%
- Bayes' theorem:
- $p(H|D) = p(H)p(D|H) / p(D)$
- Hence, $p(+|dis) = p(dis)p(+|dis) / p(+)$
- $\approx (0,001)(1)/(0,05)$
- $\approx 2\%$

GIVEN TO STAFF AND 4-YEAR HARVARD MEDICAL SCHOOL

- Disease Y = Abstract/probabilistic version:
 - Only 18% correct response ($\approx 2\%$);
 - 45% completely ignored the base-rate information and concluded 95%.
- Disease X = Frequentist version:
 - 76% gave correct response
- Frequentist representations activate mechanisms that produce bayesian reasoning.

DISEASE X

1 out of every 1000 Americans has disease X. A test has been developed to detect when a person has disease X. Every time the test is given to a person who has the disease, the test comes out positive. But sometimes the test also comes out positive when it is given to a person who is completely healthy. Specifically, out of every 1000 people who are perfectly healthy, 50 of them test positive for the disease.

Imagine that we have assembled a random sample of 1000 Americans. Given the information given above: *How many people who test positive for the disease will actually have the disease? __ out of __*

WHICH TYPE OF FOUR-LETTER WORD HAS MORE MEMBERS?

- Words that begin with an R?
- Words that have R in the third position.
- Correct answer:
 - __ R _ words are more common than R _ _ _ words
- Results
 - 69% of subjects believed that there are more R _ _ _ words
 - (averaged over 5 letters for which more words begin with the letter in the 3rd position than the 1st)
- Explanation
 - it's easier to generate R _ _ _ words than __ R _ words from memory
 - people estimate probabilities based on what they can retrieve
 - the easier it is to retrieve instances of something, the higher its estimated probability
 - often memory mechanisms do track frequency well

AVAILABILITY HEURISTIC

- A person is said to use the availability heuristic:
 - whenever he/she estimates frequency or probability
 - by the ease with which instances or associations could be brought to mind.
- To assess availability it is not necessary to perform the actual operations of retrieval or construction:
 - It suffices to assess the ease with which these operations could be performed
 - Cfr: the difficulty of a puzzle or mathematical puzzle can be assessed without considering specific solutions.
- Availability is an ecologically valid clue for the judgement of frequency
 - in general, frequent events are easier to recall or imagine than infrequent ones.

FISCHOFF & BEYTH, 1975

- Based around a contemporary event:
 - President Nixon visited China and Soviet Union in 1972
- Prior to visit:
 - Rate likelihood of various events occurring during visit
 - E.g., establishing of diplomatic mission in Peking, meeting chairman Mao, ...
- Some time after visit:
 - Try to recall your earlier judgments of likelihood
 - Increase for events that had occurred, reduction for events that had not occurred
 - Stronger when interval longer

FISCHHOFF, 1982

- Experiment
 - Descriptions of not well known historical situations
 - Estimate the probability of an outcome
- Manipulation
 - Experimental group:
 - Told the "real" outcome
 - Outcome different for different groups
 - Control group: not told
- Result:
 - People estimated the probability of "known" outcomes higher
 - Specific outcome not important:
 - Told that army X had won: Probability of victory of Army X high
 - Same story, but now told Army Y had won: Probability of victory of Army Y high

HINDSIGHT BIAS

- The knew-it-all-along effect
- After an event, it is often seen as inevitable.
 - Although beforehand few people may have anticipated it.
- Availability can account for hindsight bias, as actual outcome is most available.
 - Hard to imagine other outcomes.
- Reflects our overconfidence that we could have predicted a particular outcome accurately at some point in the past.
 - = also example of reconstruction of past, so that it matches our present knowledge

BASE-RATE NEGLECT: LAWYERS VS ENGINEERS PROBLEM

- Battery of tests/interviews with 30 (70) engineers and 70 (30) lawyers
- Thumbnail descriptions of these 30 (70) engineers and 70 (30) lawyers
- Two descriptions selected at random and indicate your probability that the person is an engineer, on a scale from 0 to 100

DICK

- Dick is a 30-year-old man. He is married with no children. A man of high ability and high motivation, he promises to be quite successful in his field. He is well liked by his colleagues.
- What is the probability that Dick is an engineer (0-100)?
- Description is uninformative with regard to profession
- Only useful info is base-rate information
- But ignored:
In both groups around 50%

JACK

- Jack is a 45-year old man. He is married and has four children. He is generally conservative, careful and ambitious. He shows no interest in political and social issues and spends most of his free time on his many hobbies which include home carpentry, sailing, and mathematical puzzles.
- What is the probability that Jack is an engineer (0-100)?
- Probability quite high.
- But, almost no effect of different cover stories = neglect of base-rate information

NO COMPLETE INSENSITIVENESS TO BASE-RATE INFORMATION

- *Suppose you are given no info whatsoever about an individual chosen at random from the sample. The probability that this man is one of the 30 (70) engineers in the sample of 100 is ___%*
- In this case, complete reliance on base-rate
- Hence
 - when “no info” is given, prior probabilities are utilized.
 - when “worthless info” is given, prior probabilities may be largely ignored and people respond as if there were no basis for assuming differences.

BASE RATE NEGLECT

- = tendency to ignore the base probabilities for each outcome
- Kahneman & Tversky:
 - representativeness heuristic explains base rate neglect.
- People assess the extent to which the description of Jack is similar to (or representative of) each of the two categories
 - To the extent that Jack is more similar to the stereotypical engineer, he is more likely to be judged an engineer.
- This assessment of similarity is independent of the prevalence of lawyers and engineers in population:
 - the resulting probability judgment is independent of the base rates for the two professions.

BASIC EXAMPLE KAHNEMAN AND TVERSKY (1972)

- “All families of 6 children in a city were surveyed. In 72 families the exact order of births of boys and girls was G B G B B G.
- What is your estimate of the number of families surveyed in which the exact order of births was B G B B B B?”
- Results:
 - 75 out of 92 participants judged B G B B B B to be less likely than G B G B B G.
- Of course:
 - Both sequences are about equally likely.
 - However, the latter is not representative of the proportion of boys and girls in the population, so it is judged less likely.

REPRESENTATIVENESS HEURISTIC

- When assessing the probability of an outcome
 - (for example, how likely it is that a person belongs to a particular category or the probability that an observed sample was drawn from a particular population)
 - people assess the *similarity between the outcome and the category (or between the sample and the population)*.
- More generally, using the representativeness heuristic involves
 - “an assessment of the degree of correspondence between a sample and a population, an instance and a category, and act and an actor or, more generally, between an outcome and a model.” (Tversky & Kahneman, 1983, p. 295).

CONJUNCTION PROBLEM = LINDA-PROBLEM

- Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.
- Please rank the following statements by their probability, using 1 for the most probable and 3 for the least probable.
 - a. Linda is a teacher in an elementary school.
 - b. Linda is a bank teller.
 - c. Linda is a bank teller and is active in the feminist movement.

CONJUNCTION PROBLEM = LINDA-PROBLEM

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- Please rank the following statements by their probability, using 1 for the most probable and 8 for the least probable.
 - a. Linda is a teacher in an elementary school.
 - b. Linda works in a bookstore and takes Yoga classes.
 - c. Linda is active in the feminist movement
 - d. Linda is a psychiatric social worker.
 - e. Linda is a member of the League of Woman Voters.
 - f. Linda is a bank teller.
 - g. Linda is an insurance sales person
 - h. Linda is a bank teller and is active in the feminist movement.

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WHY INTERESTED IN (B) AND (C)?

- (b) *Linda is a bank teller.*
- (c) *Linda is a bank teller and is active in the feminist movement.*
- (c) cannot happen unless (b) happens
- However, in T&K experiment:
 - 89% naive subjects judged (c) more probable than (b)
 - Graduate Stanford students: 85%

- Tversky and Kahneman argue that this conjunction effect arises:
 - Because people base their probability judgments on representativeness (or similarity).
- The description of Linda was intended to be:
 - representative of (i.e., similar to the stereotype of) an active feminist
 - and unrepresentative of a bank teller.
- Linda is judged more likely to be a “feminist bank teller” than a “bank teller”:
 - the addition of “feminist” means that the description of Linda is more similar to (representative of) that category.

1 X 2 X 3 X 4 X 5 X 6 X 7 X 8 VS
8 X 7 X 6 X 5 X 4 X 3 X 2 X 1

- Results
 - the median estimate of the subjects for the product on top was 512
 - the median estimate of the subjects for the product on the bottom was 2,250
 - the correct answer for both is 40,320
- Explanation
 - the first number encountered anchors the estimate
 - encountering a small initial number produces a low anchor and subsequent estimate
 - encountering a large initial number produces a higher anchor and subsequent estimate
- Application
 - sales people capitalize on this all the time
 - set a high price and then come down
 - buyers do the opposite
 - set a low price and then come up

TVERSKY AND KAHNEMAN (1974)

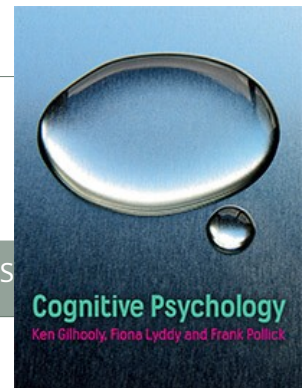
- Judge the proportion of African countries in the United Nations.
- An anchor was selected by spinning a wheel of fortune:
 - landed on 10 (low anchor) or 65 (high anchor)
- Results:
 - People given the low anchor produced smaller judgments for the percentage of African countries in the UN than those given a high anchor
 - medians = 25% and 45%, respectively.

ANCHORING

- General idea:
 - People making difficult or long sequences of judgments often use the previous item as a point of reference.
- Judgments may assimilate towards externally-provided or internally-generated *anchor values*.
- The precise effects of anchors and reference points depend on how people approach the comparison process.
- When faced with a difficult or a long sequence of difficult judgments:
 - the previous item in the series often serves as the basis for judgment of the current stimulus.

2. DECISION MAKING

SOME THEORETICAL FRAMEWORKS



UTILITY AND PROSPECT THEORY

- Subjective value or utility of a given additional amount of money decreases the more money you already have.
- An extra £1 is worth more to a penniless person than it is to a billionaire.
- Prospect Theory
 - Expected value involves subjective probability
 - People's perceptions of probability systematically depart from objective values.
 - The value function for losses is different than the value function for gains. The value function for losses is convex and relatively steep; the one for gains is concave and not quite so steep.
 - Consequence: losses "loom larger" than gains.
 - For instance, a loss of \$500 is felt more than a gain of \$500.'
 - People tend to *overweight* small probabilities (e.g., doing the lottery) and *underweight* large probabilities (e.g., not taking a chance when the odds are favourable).
 - Unlike expected utility theory, prospect theory predicts that preferences will depend on how a problem is framed:
 - If the reference point is defined such that an outcome is viewed as a gain, then the resulting value function will be concave and decision makers will tend to be risk averse.
 - If the reference point is defined such that an outcome is viewed as a loss, then the value function will be convex and decision makers will be risk seeking.
 - ...

UTILITY AND PROSPECT THEORY

- Formalized theory:

$$U(p) := \int_{-\infty}^0 v(x) \frac{d}{dx}(w(F(x))) dx + \int_0^{+\infty} v(x) \frac{d}{dx}(-w(1 - F(x))) dx,$$

TWO-SYSTEM APPROACHES TO DECISION MAKING

- The two systems framework can also be applied to decision making
- System 2 will be more involved in careful analytical decision making
 - See first slides DM
- System 1 will be more involved in decision making based on heuristics, biases and gut feelings
- Sometimes a lot (too much) focus on mistakes people make:
 - "reason is nothing more than the tail being wagged by the dog by emotions and unconscious processes"

FAST AND FRUGAL HEURISTICS: THE ADAPTIVE TOOLBOX

- Basic idea in case of choice:
 - Search for a cue that will provide a reason for choosing one alternative over the other.
 - Once such a cue found, process of searching is halted
 - The appropriate alternative is chosen
- = *also one-reason decision making*
- = *fast and frugal*:
 - Less time and less information needed
 - Because people are limited-capacity processors of information

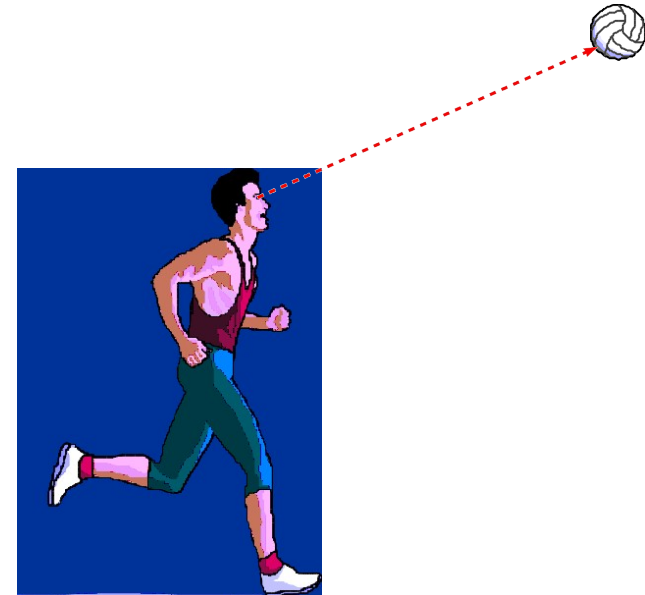
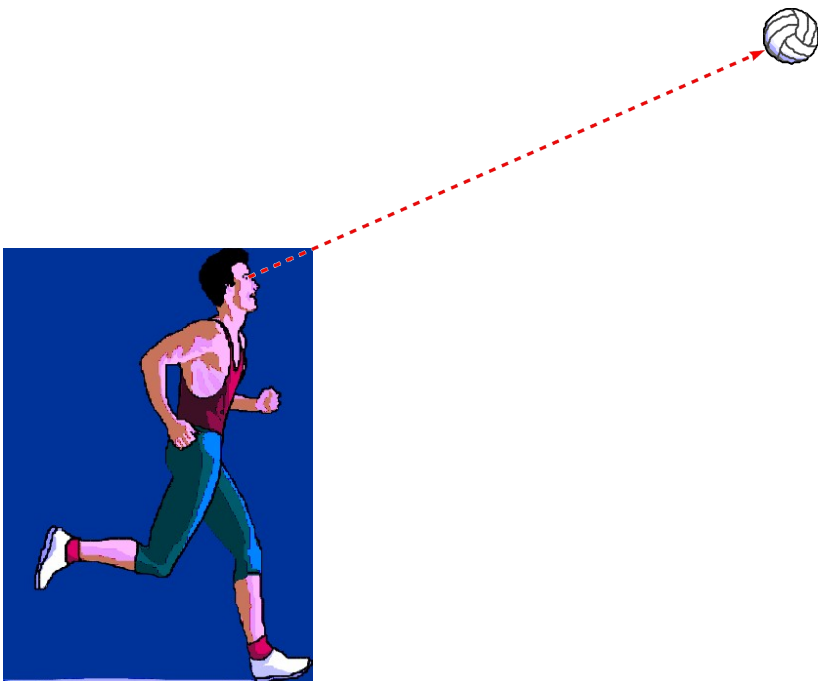
GAZE HEURISTIC

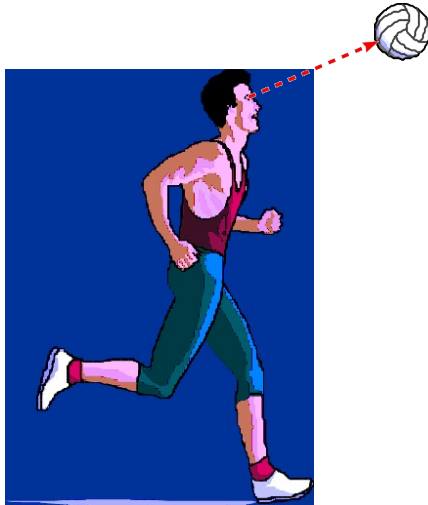
When a man throws a ball high in the air and catches it again, he behaves as if he had solved a set of differential equations in predicting the trajectory of the ball... At some subconscious level, something functionally equivalent to the mathematical calculation is going on.

Richard Dawkins, The Selfish Gene

GAZE HEURISTIC

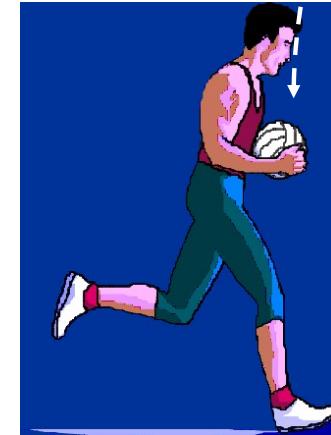
- = frugal strategy for interception of moving objects
- When ball is descending:
 - player only needs to fixate on the ball
 - adjust his running speed
 - so that the angle of gaze remains constant
- Hence, only one variable needs to be attended to





GAZE HEURISTIC

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- When ball is descending:
 - player only needs to fixate on the ball
 - adjust his running speed
 - so that the angle of gaze remains constant
- Hence, only one variable needs to be attended to
- Cfr animals



GAZE HEURISTIC

- = “fixate ball, start running, keep angle constant”
- Building block = “fixate ball”
- Evolved ability= object tracking
- When and who?
 - *Predation and pursuit*: bats, birds, dragonflies, hoverflies, teleost fish, houseflies, ...
 - *Avoiding collisions*: sailors, aircraft pilots, ...
 - *Sports*: baseball outfielders, cricket, dogs catching frisbees, ...

NOTE: Gaze heuristic ignores all causal relevant variables

THE TAKE-THE-BEST HEURISTIC: 3 STEPS

- **Search rule:**
 - Search through cues in order of their validity.
 - Look up the cue values of the cue with the highest validity first.
- **Stopping rule:**
 - If one object has a positive cue value and the other does not (0 or unknown), then stop search and proceed to next step.
 - Otherwise exclude this cue and search next one.
 - If no more cues found, guess.
- **Decision rule:**
 - = One-reason decision making
 - Predict that the object with the positive cue value has the higher value on the criterion.

THE TAKE-THE-BEST HEURISTIC: 3 STEPS

- Hence,
 - People consider cues one at a time in order of cue validity.
 - If the cue considered discriminates between stimuli, the favoured stimuli is chosen and no further cues are examined.
 - If the most valid cue does not discriminate, the next valid cue is considered and this process continues until cues are exhausted. If the cues are exhausted, select randomly.
- Or,
 - Take **the best** and ignore the rest

THE TAKE-THE-BEST HEURISTIC: 3 STEPS

- There are some variants, like the minimalist heuristic: Also 3 stages
- Search rule:
 - Select cues in a random order
- Stopping rule:
 - If one object has a positive cue value and the other does not (0 or unknown), then stop search and proceed to next step.
 - Otherwise exclude this cue and search next one.
 - If no more cues found, guess.
- Decision rule = One-reason decision making:
 - Predict that the object with the positive cue value has the higher value on the criterion.
- Hence,
 - Take **any reason** and ignore the rest.

FAST-AND-FRUGAL TREES

- FFTs are a special case of simple heuristics
- Allow efficient and effective binary classification decisions by sequentially attending to a list of diagnostic cues

FAST-AND-FRUGAL TREES

- Search rule:
 - Search through cues in a predetermined order.
- Stopping rule:
 - Stop search as soon as a cue leads to an exit.
- Decision rule:
 - Classify the object accordingly.

SEVERE CHEST PAIN?

- Patients arrive with severe chest pain:
 - acute ischemic heart disease or not
 - if so, then assigned to the intensive coronary care unit (ICU).
- Michigan hospital:
 - doctors played it safe:
 - sending about 90% of the patients to the ICU
 - Only about 25% of these actually had a myocardial infarction

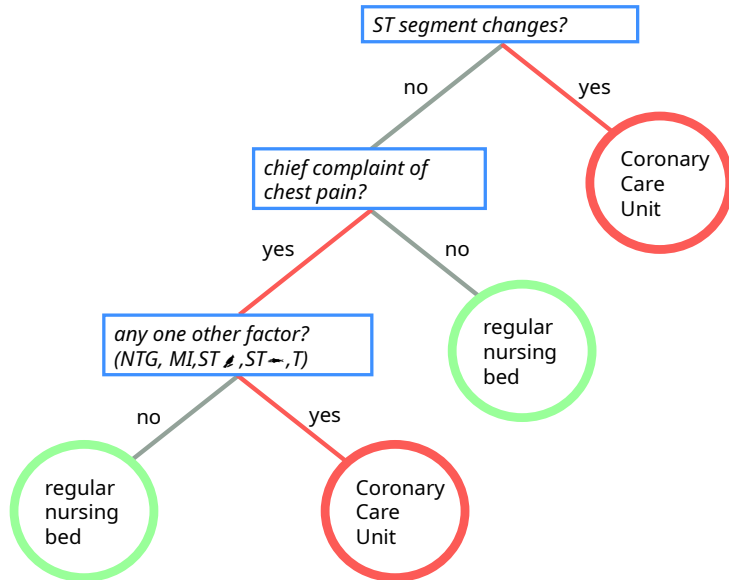
SEVERE CHEST PAIN?

- Also clear disadvantages:
 - crowded ICU, a decrease in quality of care, an increase in cost, and a risk of serious infection among those who were incorrectly assigned.
- 2 solutions:
 - a logistic regression, the Heart Disease Predictive Instrument (HDPI),
 - a fast-and-frugal tree.

SEVERE CHEST PAIN?

- HDPI:
 - = a chart with some 50 probabilities
 - Doctors checked the presence and absence of symptoms,
 - inserted the relevant probabilities into a pocket calculator

Fast-and-frugal tree



SEVERE CHEST PAIN?

- The tree was more accurate in predicting actual heart attacks than the HDPI:
 - Fewer patients who suffered from a heart attack were sent wrongly into a regular bed
 - Physicians' high false-alarm rate were nearly halved
 - Tree was transparent, easy to memorize, and easy to modify, and was accepted by physicians who disliked relying on a logistic regression they barely understood.

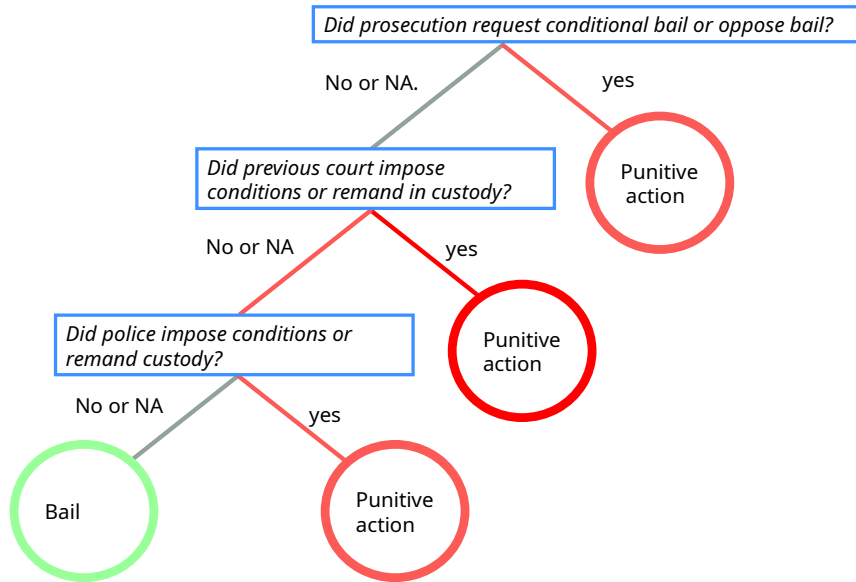
BAIL DECISIONS

- One of the initial decisions of the legal system:
 - bail a defendant unconditionally or react punitively?
- England and Wales:
 - around two million bail decisions every year by benches of two or three magistrates
 - 99.9% are members of local community without legal training.

BAIL DECISIONS

- Interviews of magistrates:
 - *"we thoroughly examine and weight all information in a complex way"*
- Dhimi (2003):
 - observed several hundreds of trials
 - average time spent on a case was 6 to 10 minutes
 - their decisions could be predicted better with a fast-and-frugal tree

Fast-and-frugal tree



BAIL DECISIONS

- Fast-and-frugal tree:
 - predicted 92% and 85% of all decisions correctly
- Weighted additive model that corresponds to what magistrates responded in the interviews:
 - Predicted 86% and 73% of all decisions correctly

BAIL DECISIONS

- Logic of the tree = "pass the buck"
 - Copies the punitive decisions of the prosecution/previous court/police
 - Ignores relevant information about the defendant

CONCLUSION

- Heuristics are not:
 - *"mental shortcuts commonly used in decision making that can lead to faulty reasoning or conclusions"*
- Research:
 - Heuristics can often be more accurate than complex "rational" strategies
- Therefore better:
 - To admit the role of heuristics

CONCLUSION

- Goal should be to formalize and understand heuristics:
 - Building blocks, core capacities, ...
 - In what environment does a given strategy (heuristic or otherwise) succeed?
- Next, their use can be effectively taught
 - which could lead to less practice variation
 - which could lead to more efficient decisions (and e.g. more efficient medical care).

CONCLUSION

- Use of simple heuristics (S1) not necessarily a disadvantage:
 - They are cheap and can be very effective.
- Simple heuristics can exploit regularities in the environment.
- Simple heuristics are maybe/probably domain-specific rather than universal strategies